Radiation Therapy Centre 2030: Personalized Self-Steering Radiotherapy

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant]*
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Planned start date of seed funding: 1-9-2022
Primary Convergence theme: Human Centered Technology and AI for Health
Secondary Convergence theme (*if applicable*): Improving Health Journeys
Tertiary Convergence theme (*if applicable*): Fundamentals of Health & Disease
Quaternary Convergence theme (*if applicable*): Technology Supported Transitions in Health(care)
Keywords: radiotherapy, personalized treatment, human-centered design, AI multi-criteria decision support, AI self-steering technology, biomarkers, health technology assessment

1.b Summary [298 words]

In over half of all cancer patients radiotherapy is indicated, often combined with other therapies[1]. Long-term side effects among increasing numbers of survivors affect patients and society[2,3]. Radiotherapy innovations have decreased side-effects, while new combination-therapies have improved outcomes [4,5]. However, these innovations impose challenges regarding accessibility, capacity planning, and financing, due to high costs and staffing shortages, hampering implementation and further development[6].

Expanding knowledge has revealed the importance of patient heterogeneity in the biology of cancer progression and treatment response[7,8]. Great promise lies in cross-disciplinary, highly-personalized cancer treatment[4,5]. However, taking patient heterogeneity into account, optimally utilizing the exponentially growing data and combination-treatment options for each patient, and optimally adapting radiotherapy during each session, prove to be extremely challenging[7].

This Convergence-consortium uniquely combines socio-economic, technical, and bio-medical scientists to realize the "Radiation Therapy Center 2030 for Personalized Self-Steering Radiotherapy". We combine a research program with a 'Living-Lab' for fast-track innovation, value optimization, and clinical validation. A Data Repository will provide access to 'real-life' clinical and 'omics' data from patients treated with photon or proton radiotherapy. We will develop image-guidance solutions to improve precision, AI-driven solutions for fast radiation-dose adaptation (self-steering), multi-criteria decision support, and biomarker assays. We ensure cost-effective innovations in line with all stakeholders' preferences by continuous validation using human-centered design techniques, health-technology assessment, and discrete-choice experiments.

This Convergence work-plan establishes a first framework, bringing the Flagship to a competitive position for future research that includes clinical intervention studies demonstrating the impact of further personalization.

The Convergence approach is crucial to achieve genuine breakthroughs in cancer treatment with radiation. With this Flagship, we can position ourselves at the forefront of development and transition to highly personalized radiotherapy in full synergy with other targeted cancer therapies, resulting in improved treatment outcomes for patients while securing affordable and accessible cancer care.

1.c Public Summary [284 words]

Meer dan de helft van alle patiënten met kanker heeft een indicatie voor radiotherapie (bestraling), vaak gecombineerd met andere therapieën. Nieuwe bestralingstechnieken hebben bijwerkingen verminderd en combinatietherapieën hebben de prognose verbeterd, maar deze vernieuwingen gaan ook gepaard met hoge prijzen, kosten en vraag naar personeel. Dit leidt tot uitdagingen voor de toegankelijkheid, capaciteitsplanning en financiering van de oncologische zorg. Daarnaast kampt een toegenomen aantal overlevenden met lange-termijn gevolgen van de behandeling, met bijbehorende consequenties voor patiënten en maatschappij.

De huidige inzichten onderstrepen het belang om rekening te houden met de diversiteit tussen patiënten, m.b.t. het biologische gedrag van kanker, de respons op de behandeling. Er wordt in de toekomst veel verwacht van gepersonaliseerde kankerbehandelingen. Echter, het nemen van de juiste behandelbeslissingen voor individuele patiënten en het toedienen van zeer precieze radiotherapie tijdens elke behandelsessie blijken grote uitdagingen. Er moet hierbij namelijk rekening worden gehouden met individuele patiëntvoorkeuren, een zeer grote toename in de hoeveelheid beschikbare data, en verschillende behandelopties voor elke patiënt.

Om gepersonaliseerde kankerbehandeling met radiotherapie (Radiotherapiecentrum 2030) te ontwikkelen is een Convergentie-consortium nodig met sociaaleconomische, technische en (bio)medische wetenschappers. Biomarker-assays zullen ontwikkeld worden die gevoeligheid op radiotherapie helpen voorspellen. AI-gestuurde multi-criteria beslissingsondersteuning zal worden ontwikkeld om samen op elk moment de juiste beslissingen te nemen. Innovaties voor betere beeldkwaliteit en AI-gestuurde snelle dosisadaptatie zullen ontwikkeld worden om de nauwkeurige zelfsturende radiotherapie mogelijk maken. Systematische evaluatie voor medische technologieën en keuze-experimenten zullen worden aangewend om kosteneffectieve toepassingen te ontwikkelen en te implementeren. Hierbij wordt rekening gehouden met patiëntpreferenties en belangen van alle betrokken partijen.

Samengevat zal dit Flagship een Convergentie-onderzoeksprogramma genereren, dat een infrastructuur voor versnelde MedTech ontwikkeling combineert met waarde-optimalisatie en validatie van gepersonaliseerde zelfsturende radiotherapie in synergie met andere gerichte kankerbehandelingen.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s) [711 words]

Contribution to the Convergence Health & Technology mission

Cancer is a leading cause of death and its incidence is rising. In over half of all cancer patients, radiotherapy is indicated, often combined with other cancer-targeting therapies. As more patients survive cancer, long-term treatment-related side effects have an increasingly significant impact on cancer survivors and society. At the same time, new innovative treatment options that may improve outcomes impose challenges to our healthcare system and society regarding accessibility (due to high costs and staffing shortages), capacity planning, and financing.

Currently, radiotherapy is still largely a static one-size-fits-all treatment, that is neither optimized over the treatment course, nor for combined use with targeted drugs. A great future promise to improve treatment outcomes lies in the personalization of radiotherapy and its optimized interaction with other targeted treatments. To realize this potential of **fully personalized radiotherapy** we need to consider patient preferences, leverage the exponentially growing 'omics' data and treatment options available for each individual patient, and deliver radiation therapy with optimal precision. The latter will be achieved by continuously adapting to changes in patient and tumor characteristics without constant human interaction – which is in other words: *self-steering.* Addressing these combined needs, requires a **strong Convergence of socio-economic, technical, and bio-medical scientists.** To this end, this RTC2030 Flagship creates a research program combining a 'Living-Lab' infrastructure for fast-track innovation with value optimization and clinical validation, in full synergy with other targeted cancer therapies.

With our convergence approach, we are able to develop the necessary technological solutions and ensure that these are most **beneficial for everyone in our society**. This is in line with the political and social desire to carefully test the introduction of new medical technology for value and cost-effectiveness. Health Technology Assessment (HTA) in its broadest sense (i.e. studying clinical effectiveness, costs, organizational aspects, patient value and social aspects) is performed from the start. Value-based healthcare (VBHC) and cost-effectiveness analyses (CEA) are used complementary; CEA on the societal level to evaluate new technologies for reimbursement decisions and VBHC on the patient and organizational levels to develop shared decision-making and stimulate continuous improvement in healthcare. This will help to allocate healthcare resources most optimally but also enable the further refinement of HTA methodology to incorporate comprehensive outcome measures. Implementing HTA in this flagship guarantees maximization of health outcomes in line with patient preferences and at affordable costs. Finally, the broad HTA approach also enables to consider equality in healthcare. In the Rotterdam-Delft region we have a very heterogeneous patient population with large differences in socio-economic status and preferences. The RTC2030 Flagship will study at an early stage to what extent implementation of personalized and self-steering radiotherapy can contribute to reducing health inequalities, fitting within the joint mission of **socio-economic equality in healthcare**.

We are convinced that the RTC2030 research program is crucial to improve health outcomes of patients undergoing radiotherapy towards 2030 and beyond. Personalized and self-steering radiotherapy are pivotal developments in the field that closely align with the H&T joint missions **of**

individual-tailored precise medical treatment for all and directly contribute to **improved life-long health and well-being for all**.

Contribution to specific Convergence Health & Technology themes
We will converge medical-technological developments in the field of radiotherapy, imaging, biomarkers, AI, and decision support with developments in the field of health-technology assessment, including value-based healthcare and CEA, and human-centered design. Thereby integrating many data sources that will contribute to a more personalized, more effective and more affordable radiotherapy treatment. Doing this, we focus on the whole patient journey. This overall ambition aligns perfectly with the **Human-centered technology and AI for Health** theme, and the **Improving health journeys** theme of Convergence Health & Technology.
We will furthermore make progress on better understanding the biological basis for personalized radiotherapy and utilizing these insights for more personalized clinical decision-making, based on the features of the individual tumor and patient. Thus, we open up new avenues for clinical diagnostic assays and actionable treatment modalities. This part of our ambition aligns with the **Fundamentals of Health and Disease** theme.
Finally, as RTC2030 also contributes to the creation of a data-driven, value-based & affordable healthcare system by developing and deploying smart digital and medical-technological innovations, our ambitions also partially overlap with the goals of the **Technology-based transitions in health(care)** theme.

2.b Synergy between disciplines [544 words]

Merging approaches and insights from distinct disciplines
RTC2030 aims at establishing a Convergence research program with accompanying Living Lab and Data Repository for the technological development, value optimization, and clinical validation of personalized self-steering radiotherapy, in full synergy with other treatment modalities. To reach these ambitious goals, we need to **integrate approaches from various disciplines** across all Convergence partners. Personalized and self-steering radiotherapy requires 1) technological expertise to develop key enabling solutions that enhance image quality and precision guidance, as well as to drive multicriteria treatment optimization (TU Delft); 2) a better biological understanding, markers of radiation sensitivity and development of biomarker assays for clinical use (Erasmus MC); and medical physics expertise to integrate these elements into treatment plan adaptation (Erasmus MC, HollandPTC). However, technological solutions and biomedical insights can only contribute to personalized radiotherapy when they are in line with patient and healthcare provider preferences (EUR, TU Delft), can be implemented in clinical practice (Erasmus MC, HollandPTC), and are affordable (EUR) and usable (TU Delft). In this Flagship we combine researchers from different disciplines and apply the Convergence approach to develop solutions for the challenges described in section 2a. The track-record and expertise of the individual team members are listed in section 4a&b below. Erasmus MC's and HollandPTC's public-private partnerships with Elekta and Varian/Siemens Healthineers will play an important role in bringing technological solutions of the RTC2030 program to clinical practice and to the market.

To reach our ambitions we also need to **integrate our scientific infrastructures**. RTC2030 proposes a 'Living Lab' model that will utilize existing facilities, including a treatment machine for external beam radiotherapy at Erasmus MC that will be made available for technological research, a proton beam line at HollandPTC, a range of imaging devices including one of the world's first clinical photon-counting CT scanners, a range of medical technology laboratories, extensive data processing infrastructure, and the new DelftBlue supercomputer of TU Delft allowing state-of-the-art algorithmic and AI research. FAIR data management will be assured by establishing a dedicated Data Repository. State-of-the-art health technology assessment methods, developed at EUR, will be applied from the beginning and in a continuous feedback loop, to ensure alignment of technological and clinical developments with the patients' needs and preferences as well as with societal demands (e.g. affordability, accessibility). Furthermore, applying assessment methods and AI in this Convergence program facilitates refinement of these methods for future research (e.g. development of new technologies and personalized treatment in other diseases).

The **uniqueness of RTC2030** lies in this strong transdisciplinary partnership establishing a Living Lab and accompanying Data Repository by (1) setting up a dedicated AI-driven data processing hub in close connection to the clinical environment and biomedical research facilities of Erasmus MC, making it possible to quickly develop, apply, assess, and validate new concepts in multi-criteria decision making and treatment automation; (2) integrating X-ray based image-guidance innovations with a dedicated research radiotherapy treatment machine, ultimately extending the photon-counting imaging techniques available in the pre-irradiation stage to the treatment room; and (3) coupling a broad human-centered, value-based approach to health technology assessment enabling validation and continuous optimization of innovations. Such an integrated and truly convergent approach is not only unique worldwide, but also necessary to sustainably improve patient outcomes by bringing affordable personalized self-steering radiotherapy to clinical practice.

2.c Long term sustainability plan [587 words]

Scientific sustainability

The RTC2030 program outlined below is necessary to improve outcomes of individual patients undergoing radiotherapy towards 2030 and beyond, while keeping our healthcare system sustainable and accessible for all. The proposed developments must be set in motion as soon as possible to address this urgent societal need. The only question is whether we will take the lead or whether our competitors will. Acquiring the Convergence Flagship funding provides us with an important impulse, enabling us to visibly integrate and intensify our efforts, to further build up our collaborative socio-economic and (bio)medical-technical approaches, acquire first results, and to integrate new research dimensions – research dimensions which are required to successfully implement the first results on the one hand, and generate fundamental (technical, radiobiological, clinical, and healthcare systems) insights to enable future progress on the other. As these scientific and medical-technological developments will take more than five years to reach full implementation, the consortium partners are already committed to an endeavor which spans beyond this period. Many of the consortium members have been at the forefront of scientific developments in their respective fields for quite a while, which means they can adapt to developments in these fields as well. By paying special attention to the training of the next generation of early-career researchers, we make the RTC2030 program durable and, over time, less dependent on its founding partners.

Financial sustainability

Collectively and often already within Erasmus MC/TU Delft/EUR collaborations, the RTC2030 members acquired substantial research funding, often as part of public-private partnerships. This already is a strong indicator of the financial viability of the RTC2030 collaboration in terms of attracting public and private funding. As the consortium aims to intensify collaborations and specifically mentor early-career researchers, we expect the number of individual and collaborative grant applications for realizing specific aims of the RTC2030 program to increase as well, with our jointly appointed post-docs being among the main drivers. As the program also explicitly is constructed to move medical-technological developments along the TRL-timeline towards valorization, we similarly expect to see increasing attention of enterprises for collaborative projects or licensing of technology. Moreover, we will acquire further funding by utilizing the project results and establish a sustainable research program on the value-based and human-centered development of the RTC2030.

Risks

The successful clinical introduction of new solutions resulting from the RTC2030 innovation program requires appropriate reimbursement by healthcare insurers and without continuous assessment from different perspectives there is a risk to deliver solutions which are not cost-effective or not in line with patient values. Providing the data needed for evidence-based reimbursement decisions therefore is an integral part of RTC2030. To make sure that only cost-effective solutions are pursued towards clinical implementation, health technology assessment will be performed from the earliest stages of development.

Parts of the new technology and methodology will be developed in-house and, if successful, can be implemented in clinical practice without the intervention of medical device manufacturers. This means that we need to adhere to the Medical Device Regulations. MDR certification will therefore be an integral part of the program. Medical physics experts will be involved to ensure proper implementation of new devices (including software) according to regulations set by the Health and Youth Care Inspectorate (Covenant on Medical Technologies).

Any medical innovation runs the risk of not being adapted by its users. To counter this risk, a thorough investigation of the needs of healthcare professionals and patients is started at the beginning of the program, combined with continuous monitoring whether our envisioned solutions truly address these needs.

2.d Visibility, attracting talents and new partners [338 words]

To ensure the **visibility** of the RTC2030 Flagship, we use targeted communication towards our stakeholders (patients and patient organizations, healthcare professionals, policy makers, citizens and health insurers) through our flagship website and social media channels. We will invite stakeholders to participate in interactive (propositions, live discussions, brainstorm sessions) cross-disciplinary meetings. During these meetings we facilitate speed-dates between our stakeholders to expand convergence. For example, industry can meet with policymakers and clinicians or patients can meet with technicians, health Insurers and biologists. Scientific results will also be disseminated using the academic channels of conferences and journal publications.

Key in our strategy to **attract talents** is the RTC2030 **Living Lab** approach. This shared infrastructure where ideas can be tested and results be implemented in virtual clinical scenario's, while FAIR data management is assured through the accompanying Data Repository, provides an ideal and attractive environment for BA/BSc and MA/MSc students from all partnering institutes to engage in scientific project as part of their training. Our research program, as outlined in section 3, offers numerous opportunities for student engagement, via contribution to subprojects on topics such as: patient journeys and workflows for use cases, interface design for technologies (WP1); image guidance in radiotherapy (WP2); experimental projects involving real radiotherapy machinery (WP3); AI-based bioinformatics (WP4); machine learning; stakeholder needs and preferences

(WP5); HTA of personalized radiotherapy (WP6). These projects are suitable for BsC/BA and MSc/MA students in e.g. Medicine, Applied Physics, Electrical Engineering (Signals and Systems), Clinical Technology, Technical Medicine, Biomedical Engineering (Medical Physics), Health Economics, Policy and Law. Moreover, the RTC2030 Living Lab can be used for training medical physicists (KLIFIO's), medical physics engineers (KFM-ers), Radiation Therapists (RTT's) and Residents in Radiotherapy.

The Living Lab approach and the broader RTC2030 ecosystem, including the clinical partners, current private partners, and stakeholders, will enhance our ability to **attract public and private funding and new partners**. Our ecosystem will be a strong source of multi-disciplinary scientific and technical innovations and therefore a highly attractive place to invest in for funding organizations and private partners alike.

2.e Long term societal impact [281 words]

The long-term aim of the RTC2030 program is to develop and clinically implement fully personalized and self-steering radiotherapy. The research program and Living Lab infrastructure are expected to result in: 1) **technical innovations** related to precision image guidance and AI-driven solutions that will primarily be commercialized by our industrial partners; 2) radiobiological insights that provide new opportunities for actionable therapies, **new biomarkers** to predict (radio)therapy sensitivity and accompanying assays suitable for routing towards clinical use; 3) the results of broad early adoption of health technology assessment and value-based healthcare will provide scientific **evidence for the development and implementation of cost-effective** innovations, contributing to a sustainable health-care; 4) actively involving patients and citizens as stakeholders throughout the development and implementation process and including socio-economic status in the analyses will ensure that the proposed solutions pay attention to inequality and will **benefit all patients**; 5) proposed labor-friendly innovations for self-steering radiotherapy, achieved through actively involving healthcare staff in development and implementation, will be essential in the near future to **maintain healthcare accessibility** in the light of increasing shortages in staffing; 6) taken together, the results will contribute to **more cure** with **less** treatment-related **side effects,** efficient and **cost-effective healthcare** in line with **patient preferences** and needs which has a direct long-term impact on society.

RTC2030 will first focus on the treatment of **patients with breast, lung and head-and-neck cancer**, together comprising approximately one-third of the 6000 patients treated each year at Erasmus MC's department of Radiotherapy. After demonstrating success for these indications, our innovations can be further expanded to other disease sites. Similarly, our iterative approach for continuous health technology assessment during technological development and implementation could be expanded to other MedTech areas.

Health & Technology
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3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions [146 words]

The long-term ambition of RTC2030 is to demonstrate that **personalized and self-steering radiotherapy has unmatched potential to improve cure and decrease side effects, while reducing treatment burden, workload, and costs**, both from a patient and societal perspective. Steps to personalize radiotherapy, e.g. introducing biomarkers for selecting adjuvant targeted therapy or modification of radiotherapy dose, require carefully planned clinical studies to provide high-level evidence of efficacy. The RTC2030 framework, established during the first 5 years using real-world clinical data, will provide essential benchmark models for such studies and to demonstrate efficacy from a clinical and societal perspective. Moreover, RTC2030 converges innovations relevant for further personalization of radiotherapy: improvements in image-guidance, fast radiotherapy dose adaptation, biomarkers related to sensitivity of normal and tumor tissue to radiotherapy, and multicriteria decision making. The Flagship combines a research program with a 'Living-Lab' infrastructure to fast-track innovation, value optimization, and clinical validation.

3.b Aims and feasibility [328 words]

Specific aims of RTC2030

The RTC2030 consortium members have set the following specific aims during the initial 5-year period (detailed under 3.c):

1. Engage all stakeholders and take a social-technical system perspective throughout the RTC2030 R&D program to make sure humane and societal needs and preferences are addressed in our solutions;
2. Develop and integrate cost-effective image guidance solutions to enable self-steering radiotherapy;
3. Develop and clinically implement self-steering radiation delivery through AI-driven methods which take into account all available data before and during treatment;
4. Identify biomarkers for individual radiation sensitivity and develop methods to utilize these for improving decision making in clinical care;
5. Develop and optimize AI-driven multicriteria treatment decision support for full personalization of the patient journey;
6. Develop models to validate personalized, self-steering radiotherapy and ensure optimal resource allocation to maximize (health) outcomes in line with patient value.

Expected results within the first five years

RTC2030 is considered successful when during the next 5-years:

1. We establish a first framework for personalized radiotherapy as described in 3c. This provides a benchmark for future clinical studies for efficacy testing of steps to further personalize radiotherapy. These future clinical studies can be operational by 2030.
2. The 'Living-Lab' infrastructure is fully functional, to ensure close collaboration within the consortium and integration of future students and researchers.
3. The aims of the individual WP's described under 3c are met, with accompanying scientific output.

4. The consortium is successful in securing continuous external funding with convergent proposals.

Long-term plan
The above results will provide the foundation for establishing a sustainable, externally-funded research program aimed at realizing RTC2030's long-term ambitions. Our jointly appointed post-docs will be important drivers to secure external funding in a convergent way, together with tenured staff from the consortium. We will create critical research mass by appointing at least 12 PhD students and post-docs in the first 5 years. Our long-term ambition is to grow RTC2030 into a comprehensive program accommodating 30-50 PhD students.

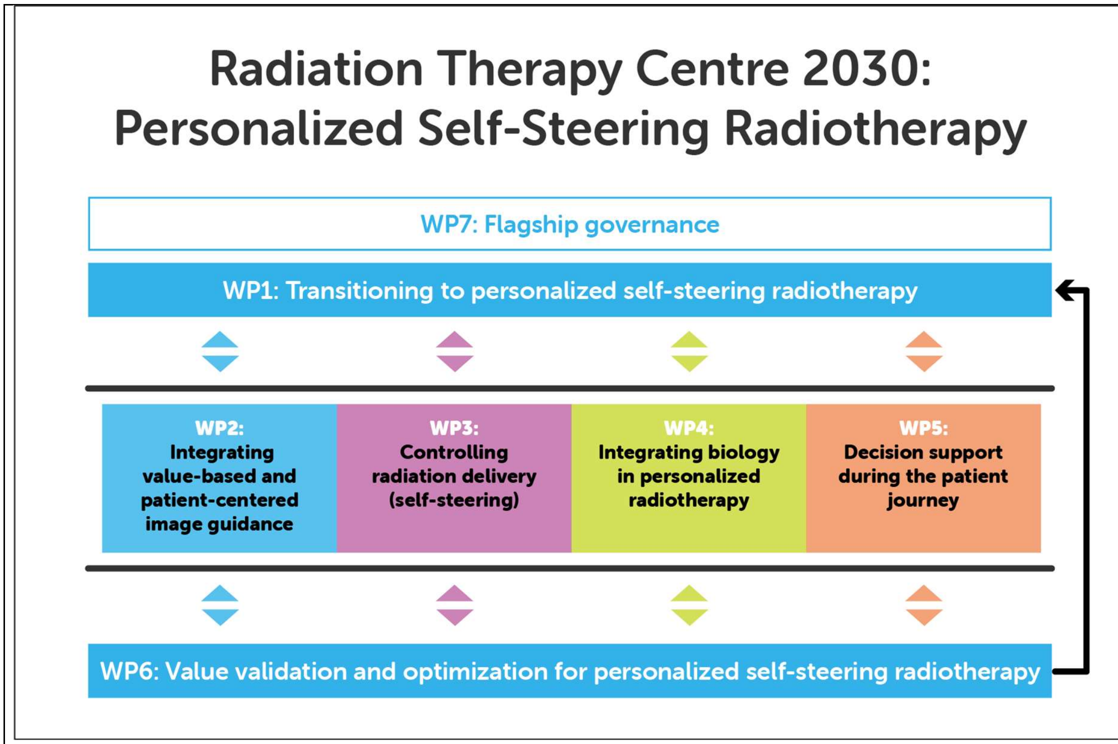
3.c Work plan [1282 words excl. figure (49 words)]

The work will be centered around **three clinical use cases**, lung, head-and-neck, and breast cancer. These sites were carefully selected based on patient volume, multimodality treatment approach, availability of treatment outcome databases, and running clinical studies including those on biological response assessment through ex-vivo assays, imaging biomarkers and cross-modality predictive modelling of treatment outcomes, including methods to identify and address bias. Moreover, selected patients of these sites are treated with protons at HollandPTC.

Work Packages and their coherence
The specific aims above correspond with the first six work packages (WPs) in the work plan, with a seventh WP for the Flagship governance. WP1 and WP6 are complementary and span the whole **transition towards personalized self-steering radiotherapy**: WP1 enables this transition by investigating and integrating **stakeholder needs and preferences** from a human-centered design perspective, while WP6 ensures that RTC2030 delivers **optimal societal value**. WP2 (image guidance) and WP4 (biological therapeutic markers) deliver key elements to **enable the self-steering and personalized** aspects of RTC2030, which is **integrated on a workflow level** in WP3. WP5 **converges these results on the treatment decision level**, by developing a clinically-usable, patient-friendly decision-support tool to guide the patient journey.

The **RTC2030 Living Lab** will provide a hybrid research and education environment encompassing physical and virtual, distributed research infrastructure. The Living Lab provides a **human-centered, co-creation approach integrating research and innovation processes in real-life hospital settings**, where researchers, students and stakeholders can connect, to design and test new technical developments, models, and algorithms, and where student projects can take place.

The **RTC2030 Data Repository** will include real-world, value-based-healthcare, imaging, lab, tissue, treatment and outcome data, as well as (technical) treatment parameters. Standardized and automated workflows recently developed at Erasmus MC together with ChipSoft, the manufacturer of the Electronic Health Record system (HiX), allow this. Accessibility of all technical data is achieved by connecting relevant digital systems. Regulatory compliance is governed by informed-consent procedures.



The work-plans of the WPs are outlined below:

WP1: Transitioning to personalized self-steering radiotherapy

WP1 will actively engage stakeholders (patients, citizens, healthcare professionals, societal partners) throughout the development of RTC2030. Identifying and integrating their needs from a human-centered systemic design perspective will ensure that the work done in WP2-5 is targeted towards effective, useful, and usable solutions. Tasks of WP1:

1. Identifying **needs and preferences of healthcare professionals** and ensuring that the work of WP2&3 is aligned with these needs;
2. Identifying **needs and preferences of patients** and citizens and ensuring that the work of WP4&5 is aligned with these needs;
3. Deliver novel **social-technical system designs** including new workflows and patient journeys to WP2-5;
4. Continuously evaluate and iterate the **impact of novel designs** on stakeholders, in collaboration with WP6.

WP2: Integrating value-based and patient-centered image guidance

1. With WP1,3,5, assess **unmet imaging needs** for radiotherapy tasks that rely on image guidance (treatment planning, adaptation, patient positioning, dose delivery, QA).
2. Early **cost-effectiveness analysis (CEA)** of different radiotherapy imaging solutions (with WP6).
3. Equip Living Lab radiotherapy gantry with a photon-counting detector (like [9]) and **demonstrate the added value of cone-beam spectral CT (CBSCT) guidance** (with WP3).
4. Assess **clinical and cost-effectiveness of photon-counting CT (PCCT)** in the pre-irradiation stage of the radiotherapy workflow (with WP3,6) and provide a **gold standard** for tasks 3,6.
5. Develop ultrafast reconstruction algorithms for PCCT and CBSCT, utilizing AI methods and deterministic forward modelling, to provide **on-the-fly optimized images for WP3**.

6. Develop intelligent photon-counting X-ray detectors [15] that **enable cost-effective implementation of CBSCT** (with WP6).
7. Develop **QA methods** based on CBSCT, prompt-gamma, and PET for self-steering radiotherapy.

WP3: Controlling radiation delivery (self-steering) for photon/proton radiotherapy

1. **Assess and prioritize unmet clinical needs** and requirements with WP1,2,5 for self-steering radiotherapy through **early HTA**.
2. Develop **AI-driven methods for fully automated treatment adaptation**; provide uncertainty metrics; investigate benefit of joint optimization of multiple tasks.
3. Develop methods for **biology-based adaptation compensating for delivered dose**, patient and tumor heterogeneity and (biological) response data from WP4,5; develop methods to predict patient and cost benefits to select patients for adaptive workflows.
4. **By design include uncertainty mitigation and quality assurance (QA)** in the integration of tasks 2,3.
5. **Develop automated patient positioning and personalized devices for patient support** for do- it- yourself patient setup without masks.
6. Develop **standardized QA and commissioning procedures and training programs** for implementation of self-steering radiotherapy.

WP4: Integrating biology in personalized radiotherapy

This WP develops a framework to predict tumor and normal tissue responses to radiotherapy. The intitial focus is on DNA damage response; the focus will be expanded in future (e.g. immune-radiotherapy) based on results of task 1.

1. Assess and prioritize unmet clinical needs (WP1) and requirements (WP5&6) for biomarkers through early HTA.
2. Determine the impact of genetic variation on DNA damage response after radiotherapy, by performing polygenic risk scores (PRS), collect and analyze liquid biopsies.
3. Identify sensitive biomarkers and PRS for early classification of patients' individual radiotherapy sensitivity; and develop a bioinformatics methodology framework, also linking radiomics (WP2,3) to inform personalized treatment decisions (WP5).
4. Develop biological assays in collaboration with WP1 for in vitro analysis, to implement biomarker / risk score testing based on the results of task 1 and 2.
5. Address the ethical, legal, and socio-economic issues related to the proposed genetic screening and biomarkers, with WP1&6.

WP5: Decision support during the treatment journey

1. **Develop a framework** (with WP1) describing decision points and informational needs of stakeholders and identify which could benefit from **treatment decision support**;
2. **Assess preferences of stakeholders** for decision support systems (e.g. informational needs, content, lay-out);
3. Enable **access to the RTC2030 Data Repository** for the algorithm driving the decision support system to:
    a. **Quantify preferences** of patients for treatments within the three focus areas (WP5),
    b. **Develop methods** to integrate:
        i. Prospectively updated clinical outcome data (e.g. value-based heath care)
        ii. Radiomics data (with WP2,3)
        iii. Bioinformatics data (with WP4)
4. **Develop backbone algorithm** of the decision support system;
5. **Develop interface** with clinicians and patient representatives;

6. **Test/Pilot** the decision support system

Qualitative interviews with **quantitative surveys** will be applied (task 1-3a and 6) and probabilistic **machine learning AI driven algorithms** will be developed to predict patient outcomes and rank potential interventions (tasks 3b-5).

WP6: Value validation and optimization for personalized self-steering radiotherapy

1. **Continuous HTA model building and validation** for use-cases to
   -assess personalized, self-steering radiotherapy
   -support evidence-based decision-making
   -ensure optimal resource allocation (efficient, equitable, sustainable).

Using data and findings from WP1to5;

2. Establish **framework** to enable future validation of personalized interventions.
3. Develop **planning methods** and **value assessment** for **photon** and **proton radiotherapy**
4. Develop **optimization methods for multimodality treatments** (e.g. radio-immuno) and perform **cost-effectiveness analysis**;
5. With WP1,5, assess and prioritize patient biomarkers (in connection to WP2to4), to improve personalized (combination) radiotherapy treatment;
6. **Train and validate prediction models using data repository** and IT-infrastructures, that continuously monitors and updates characteristics and outcomes (identified and quantified in WP1to5);
7. With WP1, identify **implementation pathways** and develop **uptake strategies** for the generated models by **key stakeholders** (patients, clinicians, policy makers).

In WP5,6 we pay attention to problems in establishing causal inference. We will design neural causal and other models to take causality into account. Methods to identify and address bias, such as directed acyclic graphs (DAGs) and propensity score matching, will be used.

WP7: Flagship governance

This WP includes all activities to ensure that the scientific and educational work of this Flagship and the societal and economic impact of its results are adequately managed (see section 4.d).

Data management plan

The RTC2030 consortium will deliver an extensive data management plan, in line with applicable regulations and the FAIR-principles, within three months after approval of the Flagship.

3.d Alignment of the program with relevant national and international research agendas [194 words]

RTC2030 is strongly aligned with the Cancer Mission recently launched by the **European Commission** with the specific objectives “Optimise diagnostics and treatment” and “Support quality of life.” Nationally, the proposal fits within the following routes of the **Dutch Research Agenda (NWA)**: “Healthcare research, sickness prevention and treatment,” “Personalised medicine: the individual at the centre” and “Measuring and detecting: anything, anytime, anywhere.” Pertinent NWA questions include: “How do we improve the quality of healthcare as much as possible while keeping it affordable?” “How can we personalise healthcare, for example by using biomarkers?” and “How do we develop minimally invasive techniques and interventions for diagnosis, prognosis, and treatment?” Moreover, the proposal is strongly linked with the **Dutch Top Sectors** “Top Sector Life Sciences & Health” and “High Tech Systems and Materials” and the respective **Knowledge and**

Innovation Agenda's (KIA) Health & Care 2020-2023 (Longer in Good Health) and Key Technologies 2020-2023 (Digital Technologies, Photonics and Light Technologies). RTC2030 furthermore ties into the focal point "Diagnostics and Minimal-Invasive Techniques" of the **Dutch Cancer Foundation (KWF)** and also addresses knowledge gaps addressed in the research agendas of the **Dutch Society of Radiation Oncology (NVRO) and Medical Physics (NVKF)**.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members [471 words]

Academic research

The RTC2030 partners each have a strong track academic record: **Erasmus MC** being home to the largest Radiotherapy department of the Netherlands with a strong innovative focus, access to renowned bio-medical and clinical research groups including the department of Molecular Genetics, Medical Oncology, Pulmonary Medicine, Surgery and a department of Radiology with clinical access to photon-counting CT; **TU Delft** with its vast medical technology and AI research and education program, **EUR** as a world-leading institute in methodological and applied research in health technology assessment, health decision making, and philosophy, and **HollandPTC** as a leading centre (and Convergence Square) for proton therapy and research.

Translation of scientific results into clinical application or societal impact

The partners have ample experience in bringing scientific results towards clinical implementation, commercialization, and other societal impact. For example, the HTA methodology developed at EUR is recommended as best practice by the National Health Care institute and included in guidelines for cost-effectiveness research. HTA-models developed at EUR are used for (inter)national reimbursement decisions (e.g. UK, Canada)[40,41]. Erasmus MC was first worldwide to clinically implement automated treatment planning, the method (Erasmus-iCycle) is currently commercialised by the major radiotherapy vendor Elekta AB (Sweden). First introduced by Erasmus MC, online adaptive radiotherapy to more accurately irradiate cervical cancer has been adopted worldwide. The REpair CAPacity, or RECAP assay, developed at Erasmus MC is a relatively easy assay for biopsy material and powerful tool for selecting patients with an HR-deficient cancer for PARP inhibitor treatment, is currently being tested in clinical trials. Seven start-up companies spun out of the Interactive Mechanisms research performed at TU Delft, including Armon Products BV (commercializing the Ayura arm support device), and Laevo BV (developing exoskeletons).

BA/BSc and MA/MSc supervision and engagement

All members have extensive experience in supervising undergraduate and graduate students. The RTC2030 Living Lab provides an excellent training ground for BA/BSc and MA/MSc students (see section 2.d above for examples).

External funding

For the next years, the following funding has been requested or awarded that contributes to the objectives of this Flagship. **Erasmus MC**: private-public partnership with Varian/Siemens Healthineers (2022-2029), NWO-Vidi (S. Breedveld) on real-time treatment plan generation (awarded; 2022-2026), various grants from KWF, EITHealth, Elekta, Accuray, NWO, Daniel den Hoed Foundation, BeterKeten. **HollandPTC-consortium**: NWA-grant “Less is more” (in preparation; 2023-2029) submitted by TU Delft, private-public partnership HollandPTC-consortium and Varian/Siemens Healthineers (awarded; 2016-2026). **EUR**: In collaboration with HollandPTC a Medical Delta grant has been awarded for HTA research and the value proposition of proton therapy at HollandPTC. Funding opportunities at ZonMW will be explored for “Efficiency in health” and “HTA Methodology”. **TU Delft**: Public-private partnerships on photon counting X-ray detectors for proton therapy, various grants from KWF, Marie-Curie, NWO-HTSM, and NWO-Veni (Z. Perko) in the field of high-precision and adaptive radiotherapy. Grants are submitted on FLASH proton therapy, proton therapy range verification.

4.b Expertise of the consortium members in relation to the aims and objectives of the program [1046 words]

WP1 – Transitioning to personalized self-steering radiotherapy		
Dr.ir. Marijke Melles (**lead**)	TUDelft	human-centered design for complex sociotechnical systems, in particular for supporting inter-professional teamwork and patient participation in care networks
Prof.dr. Marco van Vulpen (**co-lead**)	ErasmusMC, TUDelft, HollandPTC	proton therapy, new radiotherapy techniques, MRI accelerator and proton therapy, especially in the context of determining the clinical value
Prof.dr. Hub Zwart (**co-lead**)	EUR-ESPHIL	Philosophy of science, Ethics of science
Dr. Hedwig Blommestein	EUR-ESHPM	health technology assessment including economic evaluations of medical devices, experience in developing disease modelling methods for entire patient journeys from a health economic perspective
WP2 – Integrating value-based and patient-centered image guidance		
Dr.ir. Dennis Schaart (**lead**)	TUDelft	development of new technology for medical imaging and image guidance in radiotherapy
Dr.ir. Marcel van Straten (**co-lead**)	ErasmusMC	medical physics, experience in optimization of x-ray imaging techniques, especially in dual-energy and photon-counting CT
Dr. Frederick. W.F. Thielen (**co-lead**)	EUR	cost studies from a hospital and societal perspective. Experience in building health economic models for technology assessment from a societal perspective
Dr.ir. Danny Lathouwers	TUDelft	fast dose computation, stochastic optimization and image reconstruction
Dr.ir. Marlies C. Goorden	TUDelft	tomographic image reconstruction, development and optimzation of imaging systems
Prof.dr. Aad van der Lugt	ErasmusMC	Radiologist, head and neck, and neuro radiology, photon-counting CT, radiomics
Prof.dr. Marco van Vulpen	ErasmusMC, TUDelft, HollandPTC	proton therapy, new radiotherapy techniques, MRI accelerator and proton therapy, especially in the context of determining the clinical value
Dr. Steven J.M. Habraken	ErasmusMC, HollandPTC	medical physics, radiotherapy QA
Dr. Frans M. Vos	TUDelft	Medical image processing
Prof.dr. Mischa S. Hoogeman	ErasmusMC	high-precision and adaptive radiotherapy, proton therapy, prediction modelling, clinical implementation of novel technologies, medical physics
Prof.dr.ir. Alle-Jan van der Veen	TUDelft	signal processing methods and algorithms for estimation and detection, reconstruction after sparse sampling, and image-formation algorithms
WP3 – Controlling radiation delivery (self-steering)		
Prof.dr. Mischa S. Hoogeman (**lead**)	ErasmusMC, HollandPTC, TUDelft	high-precision and adaptive radiotherapy, proton therapy, prediction modelling, clinical implementation of novel technologies, medical physics

Dr. Zoltan Perko (**co-lead**)	TUDelft	biology-based radiotherapy treatment optimization, deep learning for medical applications, uncertainty quantification and numerical modelling
Dr. Ken Redekop (**co-lead**)	EUR	preclinical health technology assessment of medical tests and devices
Prof.dr.ir. Just Herder	TUDelft	mechatronic systems, interactive mechanisms, robotics, and applications of mechatronics in healthcare
Dr.ir. Danny Lathouwers	TUDelft	fast dose computation, stochastic optimization and image reconstruction
Prof.dr. Marleen de Bruijne	ErasmusMC	AI in medical image analysis, segmentation, quantitative imaging biomarkers, image-based prediction models
Dr. Marius Staring	TU Delft	Automated image segmentation
Prof.dr. Ben Heijmen	ErasmusMC	Medical physics, automated radiotherapy dose planning, clinical implementation of novel technologies
Dr. Sebastiaan Breedveld	ErasmusMC	Multicriteria decision making; optimization for automated treatment planning
Dr. Martijn Starmans	ErasmusMC	Radiomics, quantitative imaging biomarker development
Dr. Joost Nuyttens	ErasmusMC	Radiation Oncologist, lung and GI cancer, SBRT high-precision and adaptive radiotherapy, proton therapy
Dr. Steven Petit	ErasmusMC	Biological adaptive radiotherapy, workflow optimization
Prof.dr. Marco van Vulpen	ErasmusMC, TUDelft, HollandPTC	proton therapy, new radiotherapy techniques, MRI accelerator and proton therapy, especially in the context of determining the clinical value
WP4 – Integrating biology in personalized radiotherapy		
Dr. Jeroen Essers (**lead**)	ErasmusMC	molecular imaging, mouse models, understanding molecular mechanisms and biologic function in cancer, radiobiology
Dr. Hedwig Blommestein (**co-lead**)	EUR	health technology assessment including economic evaluations of medical devices, experience in developing disease modelling methods for entire patient journeys from a health economic perspective
Prof.dr. Marcel Reinders (**co-lead**)	TUDelft	Bioinformatics, machine learning, pattern recognition
Prof.dr. André Uitterlinden	ErasmusMC	Complex genetics of common diseases, DNA polymorphisms, epidemiology, aging
Prof.dr. John Martens	ErasmusMC	Translational genomics of cancer, liquid biopsies, breast cancer
Dr. Dik van Gent	ErasmusMC	DNA damage-repair in tissue and tumors, anti-tumor therapies, mechanistic studies using tumor slices, radiobiology
Prof.dr. Roland Kanaar	ErasmusMC	molecular genetics, mechanisms of DNA damage-response, and related targeting precision cancer therapy, radiobiology
Dr. Agnes de Jager	ErasmusMC	Medical Oncolgist, breast cancer and related translational research
Dr. Marta Capala	ErasmusMC	Radiation Oncologist, breast and head and neck cancer, and related radiobiological translational research
Dr. Femke Froklage	ErasmusMC	Radiation Oncologist, breast cancer, research in interaction surgery and radiotherapy, translational research
Dr. Jos Elbers	ErasmusMC	Radiation Oncologist, head and neck, and lung cancer, alternative fractionation and immune radiotherapy interaction

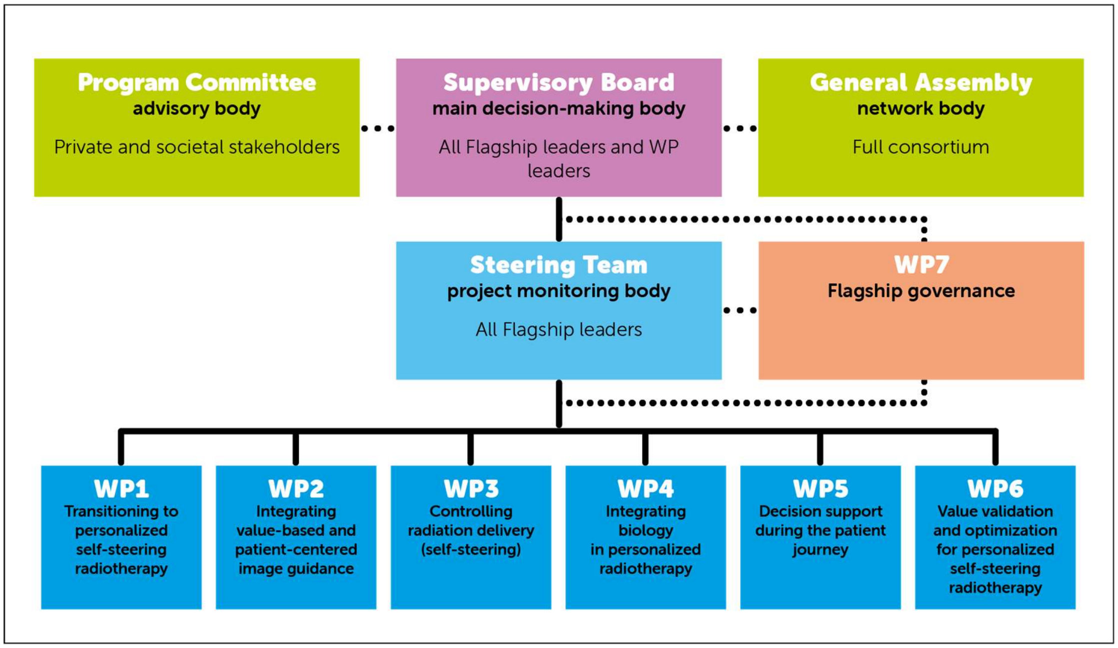
Prof.dr. Remi Nout	ErasmusMC	(adaptive) radiotherapy, gynecological cancers, clinical implementation and evaluation of new techniques, molecular risk stratification, quality of life research, image guidance
WP5 – Decision support during the patient journey		
Dr. Jorien Veldwijk (**lead**)	EUR	Discrete Choice Experiments (DCE) and obtaining preferences for and willingness to participate in different public health interventions
Dr. Justin Dauwels (**co-lead**)	TUDelft	artificial intelligence (AI) (machine learning, neural networks, graphical models), and applications of AI in healthcare
Prof.dr. Remi Nout (**co-lead**)	ErasmusMC	(adaptive) radiotherapy, gynecological cancers, clinical implementation and evaluation of new techniques, molecular risk stratification, quality of life research, image guidance
Dr. Linetta Koppert	ErasmusMC	Breast cancer surgeon, value based healthcare, shared decision making,
Dr. Femke Froklage	ErasmusMC	Radiation Oncologist, breast cancer, research in interaction surgery and radiotherapy, translational research
Prof.dr. Rob Baatenburg de Jong	ErasmusMC	Head and Neck surgeon, value based healthcare
Dr. Aniel Sewnaik	ErasmusMC	Head and Neck surgeon, value based healthcare
Dr. Marinella Offerman	ErasmusMC	Value based healthcare, use of ePROs in clinical practice, prognostic counseling, shared decision making
Dr. Jos Elbers	ErasmusMC	Radiation Oncologist, head and neck, and lung cancer, alternative fractionation and immune radiotherapy interaction
Prof.dr. Anne-Marie Dingemans	ErasmusMC	Respiratory physician, specialized in lung cancer,
Dr. Joost Nuyttens	ErasmusMC	Radiation Oncologist, lung and GI cancer, SBRT high-precision and adaptive radiotherapy, proton therapy
WP6 – Value validation and optimization for personalized self-steering radiotherapy		
Dr. Hedwig Blommestein **(lead)**	EUR	health technology assessment including economic evaluations of medical devices, experience in developing disease modelling methods for entire patient journeys from a health economic perspective
Dr. Martijn Lolkema (**co-lead**)	ErasmusMC	large scale genomic analyses of metastatic cancer tissues, translational biomarker studies in urogenital cancers and early phase drug development in oncology
Dr. Saba Hinrichs-Krapels **(co-lead)**	TUDelft	Systems and design approaches in complex decision-making environments
Dr. Zoltan Perko	TUDelft	biology-based radiotherapy treatment optimization, deep learning for medical applications, uncertainty quantification and numerical modelling
Dr.ir. Danny Lathouwers	TUDelft	fast dose computation, stochastic optimization and image reconstruction
Prof.dr. Carin Uyl-de Groot	EUR	Economic evaluations cancer treatment. Research activities focusing on rapid access to innovative treatments in the Netherlands and Europe
Dr. Marinella Offerman	ErasmusMC	Value based healthcare, use of ePROs in clinical practice, prognostic counseling, shared decision making

4.c Involvement of partners [192 words]

The RTC2030 founding partners bring their respective ecosystems into the Flagship. These partnerships will be further strengthened by Convergence, offering a more integrated approach to the challenges of cancer care. We will engage the following groups of stakeholders to co-create our research, development, and implementation plans: **Patients** – consortium members have successfully collaborated with Borstkankervereniging Nederland (BVN), Patiëntenvereniging Hoofd-Halskanker (PVHH), and Longkanker Nederland, representing the three indications we will target first. **Industrial partners -** Technological innovations, e.g. in the areas of image-guidance technology and multi-criteria decision making, will primarily be brought to the market/clinic by our industrial partners, who are leading players across the entire spectrum of medical hard- and software technology (e.g., Varian, Siemens, Philips, RaySearch, Elekta, Accuray, Varex, Broadcom, Saint-Gobain, Chipsoft). Two of them, Varian and Elekta have already indicated their interest in the Flagship Letters of Intent. **Healthcare professionals** – we will include the perspective of clinical experts and expert organization (NVKF, NVRO) as well as health insurers to ensure implementation and reimbursement of our innovations. These groups will have representatives in the Program Committee (see below), and also be consulted specifically via the stakeholder analyses in WP1, WP5, and WP6.

4.d Governance, organization and collaboration strategy of the consortium [267 words + excl. figure (76 words)]

The **governance** of the Flagship program is organized on three levels: 1) day-to-day execution, 2) co-ordination, and 3) decision-making.



On the **decision-making** level, the *Supervisory Board* with all Flagship leaders and WP leaders, is the main decision-making body. This board is advised by the *General Assembly* of all consortium members. The Supervisory Board receives independent advice from the *Program Advisory*

Committee, consisting of independent scientific experts and stakeholder representatives. This ensures that the scientific goals of the consortium remain state-of-the-art and that results of the consortium are implemented with optimal societal and economic impact. The Supervisory Board convenes once a year, together with the General Assembly and Program Advisory Committee.

The Flagship leaders together form the *Steering Team*, responsible for monitoring the project's progress and **co-ordination** between the work packages. The Steering Team meets once every two months to discuss the project's progress and identify and resolve possible obstacles. This process and related activities is prepared and supported by the Flagship leads through WP7 on project coordination.

Finally, the day-to-day **execution** of the work packages is supervised by the *Work Package Leaders* who organize regular WP meetings. All WPs involve members from the three Convergence institutes and there will be continuous **interaction** and strong **collaboration** between the WP members to reach the convergent RTC2030 goals. The RTC2030 Living Lab environment is key in our strategy to enhance the interaction and collaboration between researchers and students from different sites. The WP-members will convene regularly (at least once a month). The exact frequency and nature of these meetings will be such that the WP goals are reached most efficiently.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

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